

Calculus and Analytical Geometry

course # 355

Credit hours : 3

Books :-

- i) Calculus and Analytical Geometry by Kenneth W. Thomas.
- ii) Calculus by Stewart, and James.
- iii) Calculus by Carl William Smokowski;
Michael Binick, Dennis Pence, Jeffrey A. Cole.
- iv) Calculus by Anton Bivens Davis (10th Edition)

Calculus :- It is the study of continuous change. Issac Newton and Gottfried Wilhelm Leibniz are the pioneers of Calculus. It deals with limits, differentiation and integration of functions of one or more variables. The two major concepts of calculus are.

Derivative :- It gives us the rate of change of a function. It describes the function at a particular point.

Integrals :- It gives us the area under curve. It gathers the different the different values of a function over a number of values.

Analytical Geometry :- This also known as co-ordinate geometry or Cartesian geometry is the study of geometry using a co-ordinate system. It is a combination of algebra and geometry used to model geometric objects.

Rene Descartes (Renatus Cartesius in Latin) is the founder/father of it.

Chapter 0 "Before Calculus"

Function :- A function f is a rule that gives a unique output with each input. Input is denoted by x and output is denoted by $f(x)$. By unique it means "exactly one".

Independent and Dependent Variables:-

$$y = f(x)$$

Here the variable x is called the independent variable (or argument) of f and the variable y is called the dependent variable of f .

eg $y = 3x^2 - 4x + 2$

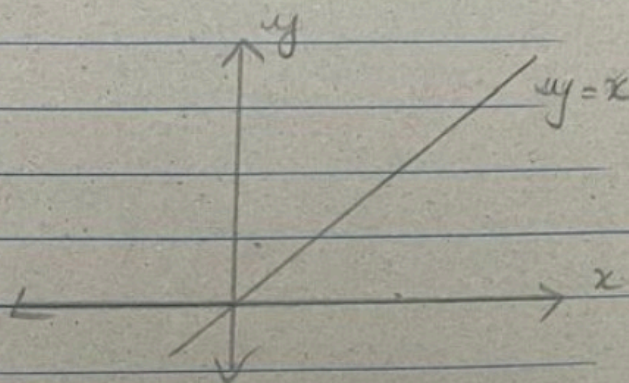
here $y = f(x)$

$$f(x) = 3x^2 - 4x + 2$$

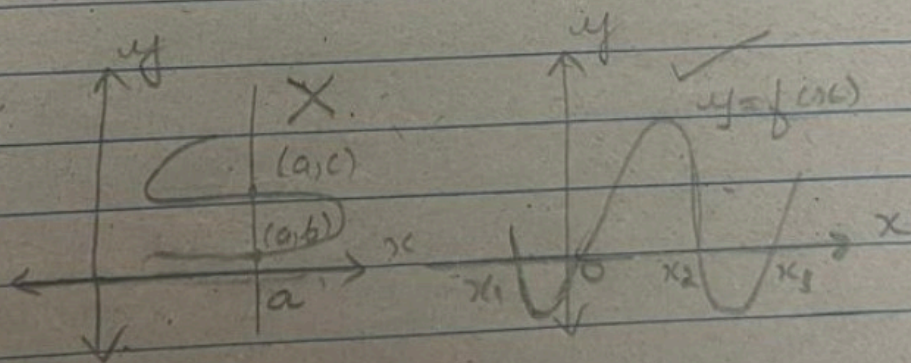
$$f(0) = 2, f(1) = 1, f(2) = 6, f(3) = 17$$

Graphs of Functions-

If f is a real-valued function of a real variable, then the graph of f in the xy -plane defined to be the graph of the equation $y = f(x)$.



The Vertical Line Test:- A curve in the xy -plane is the graph of some function f iff no vertical line intersects the curve more than once.



Absolute Value:- The absolute value $|x|$ of a real number x is the non-negative value of x without regard to its sign. It refers to the distance of a number zero and is always +ve.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Properties of Absolute Value :-

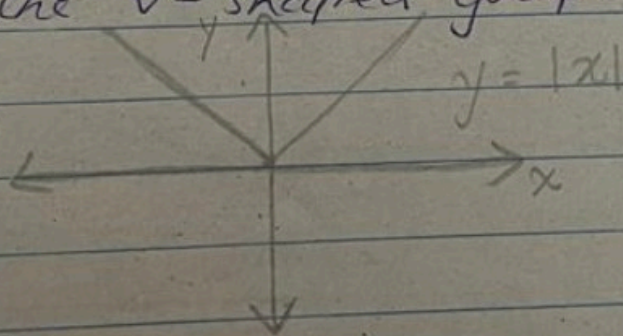
If a and b are real numbers, then

- a. $|a| = |a|$, b. $|ab| = |a||b|$ c. $|a/b| = |a|/|b|, b \neq 0$.
 d. $|a+b| \leq |a| + |b|$ the triangle inequality.

The graph of the function $y(x) = |x|$ can be obtained by graphing the two parts of the equation.
 (Note: Absolute values have important relationships to $\sqrt{\quad}$

$$y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Separately by combining the two parts produces the V-shaped graph

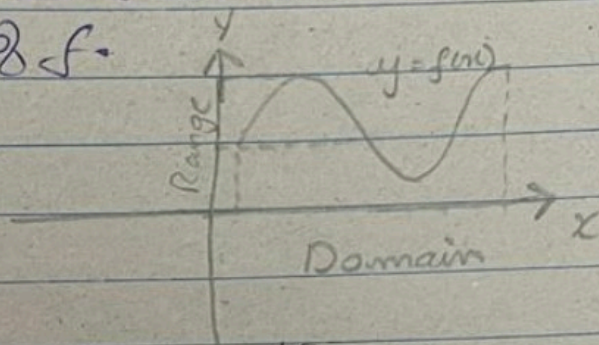


Piecewise - Defined Functions.

The absolute value function $y(x) = |x|$ is an example of a function - i.e. defined piecewise in the sense that the formula for y changes, depending on the value of x .

Domain and Range-

If x and y are related by the equation $y = f(x)$, then the set of all allowable inputs (x -values) is called the domain of f , and the set of outputs (y -values) that results when x varies over the domain is called the range of f .



Arithmetic Operations On Functions -

Given functions f and g , we define

- i) $(f+g)(x) = f(x) + g(x)$ ii) $(f-g)(x) = f(x) - g(x)$
iii) $(fg)(x) = f(x)g(x)$ iv) $(f/g)(x) = f(x)/g(x)$

For the functions $f+g$, $f-g$ and fg we define the domain to be the intersection of the domains of f and g and for the function f/g we define it to be the same but with the points where $g(x) = 0$ excluded.

Composition Of Functions - Given functions f and g , the composition of f with g , denoted by $f \circ g$, is the function defined by.

$$(f \circ g)(x) = f(g(x))$$

The domain of $g \circ g$ is defined to consist $\forall x$ in the domain of g for which $g(x)$ is in the domain of f .

Example 3: Let $f(x) = x^2 + 3$ and $g(x) = \sqrt{x}$
 find a. $(f \circ g)(x)$ b. $(g \circ f)(x)$

a. $(f \circ g)(x)$

Sol: The formula for $f(g(x))$ is

$$\begin{aligned} f(g(x)) &= [(g(x))^2 + 3] \\ &= [(\sqrt{x})^2 + 3] \end{aligned}$$

$$f(g(x)) = x + 3$$

$\therefore$ the domain of g is $[0, +\infty)$

and the domain of f is $(-\infty, +\infty)$

then the domain of $g \circ g$ is $[0, +\infty)$

$$\therefore (f \circ g)(x) = x + 3 \quad x \geq 0$$

b. $(g \circ f)(x)$

Sol: The formula for

$g(f(x))$ is

$$g(f(x)) = \sqrt{f(x)}$$

$$g(f(x)) = \sqrt{x^2 + 3}$$

$\therefore$ the domain of f is

$(-\infty, +\infty)$

and the domain of g is

$[0, +\infty)$

then the domain of

$g \circ f \quad \forall x$ is $(-\infty, +\infty)$

$$\therefore (g \circ f)(x) = \sqrt{x^2 + 3}$$

Function As A Composition :- Many problems in mathematics are solved by "decomposing" functions into compositions of simpler functions, for e.g. consider

$$h(x) = (x+1)^2$$

$$\Rightarrow h(x) = x^2 + x + 1$$

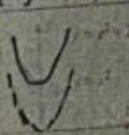
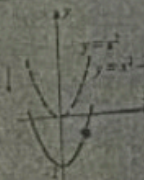
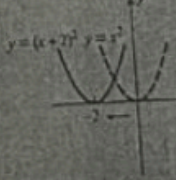
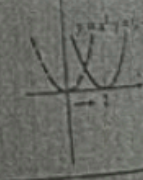
These two operations are performed by the functions $g(x) = x+1$ and $f(x) = x^2$

Now, it can be expressed by h in terms of f and g by writing

$$h(x) = (x+1)^2 = [g(x)]^2 = f(g(x))$$

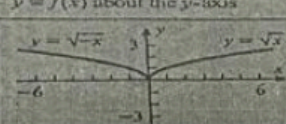
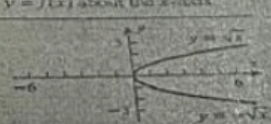
Graphical Changes :-

Translations. The geometric effect on the graph of $y = f(x)$ of adding or subtracting a positive constant c to f or to its independent variable x .

Table 0.2.2 TRANSLATION PRINCIPLES				
$y = f(x)$	Add a positive constant	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
New	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
Geometric Effect	Translates graph $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph $y = f(x)$ right c units
Example				

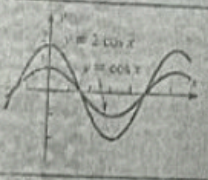
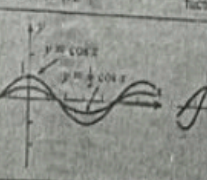
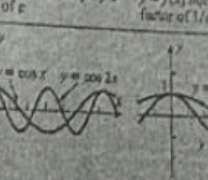
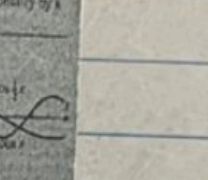
Reflections :- The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ about the y -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(-x, y)$. Similarly, the graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ about the x -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(x, -y)$. Hence

$$y = -f(x) \Leftrightarrow -y = f(x)$$

Table 0.2.3 REFLECTION PRINCIPLES		
OPERATION ON $y = f(x)$	Replace x by $-x$ $y = f(-x)$	Multiply $f(x)$ by -1 $y = -f(x)$
NEW EQUATION		
GEOMETRIC EFFECT	Reflects the graph of $y = f(x)$ about the y -axis	Reflects the graph of $y = f(x)$ about the x -axis
EXAMPLE		

Stretches and Compressions :-

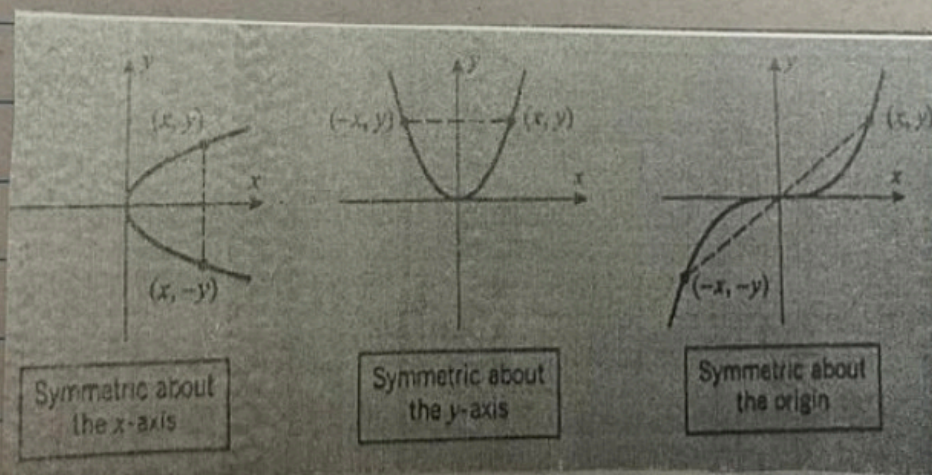
Multiplying $f(x)$ by a +ve constant c has the geometric effect of stretching the graph of $y = f(x)$ in the y direction by a factor of c if $c > 1$ and compressing it in the y -direction by a factor of $1/c$ if $0 < c < 1$.

Table 0.2.4 STRETCHING AND COMPRESSING PRINCIPLES				
OPERATION $Y=f(x)$ New Eqn.	Multiply $f(x)$ by c ($c > 1$)	Multiply $f(x)$ by c ($0 < c < 1$)	Multiply x by c ($c > 1$)	Multiply x by c ($0 < c < 1$)
GEOMETRIC EFFECT	$y = c f(x)$ Stretches the graph of $y = f(x)$ vertically by a factor of c	$y = c f(x)$ Compresses the graph of $y = f(x)$ vertically by a factor of $1/c$	$y = f(cx)$ Compresses the graph of $y = f(x)$ horizontally by a factor of c	$y = f(cx)$ Stretches the graph of $y = f(x)$ horizontally by a factor of $1/c$
EXAMPLE				

Symmetry :-

Three types of symmetries are

- Symmetry about the x -axis
- Symmetry about the y -axis
- Symmetry about the origin

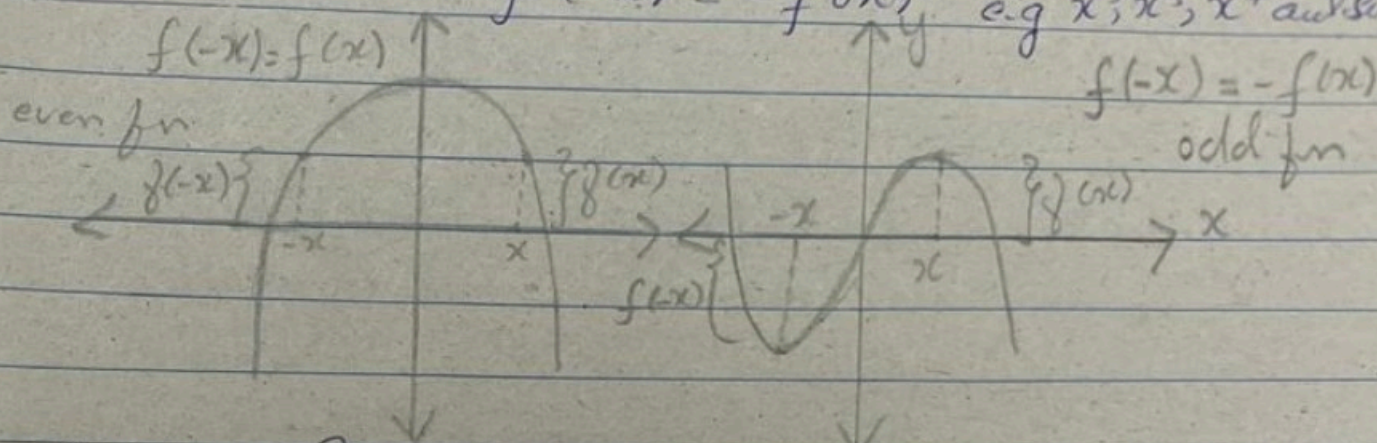


0.2.3 THEOREM (Symmetry Tests)

- A plane curve is symmetric about the y -axis if and only if replacing x by $-x$ in its equation produces an equivalent equation.
- A plane curve is symmetric about the x -axis if and only if replacing y by $-y$ in its equation produces an equivalent equation.
- A plane curve is symmetric about the origin if and only if replacing both x by $-x$ and y by $-y$ in its equation produces an equivalent equation.

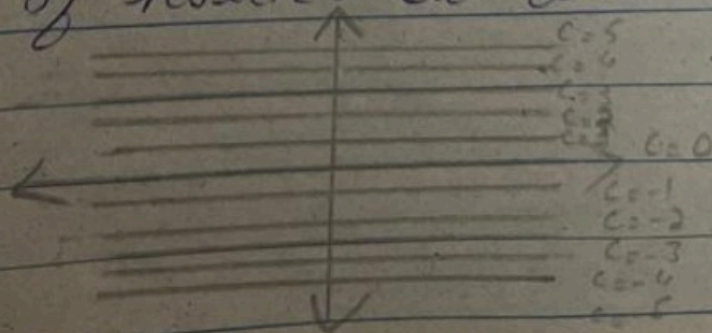
Even and Odd Functions

A function f is said to be an even function if $f(-x) = f(x)$ e.g. x^2, x^4, x^6 and $\cos x$ and is said to be an odd function if $f(-x) = -f(x)$ e.g. x^3, x^5, x^7 and $\sin x$



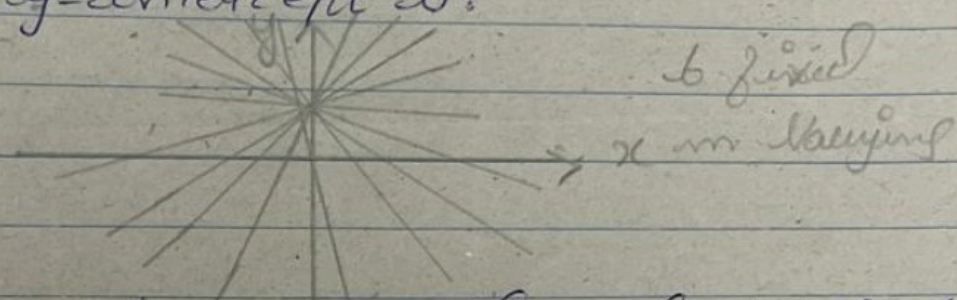
Families Of Functions - Functions are often grouped in to families according to the form of their defining formulas or other common characteristics.

Family Of Curves - The graph of a constant function $f(x) = c$ is the graph of the equation $y = c$, which is the horizontal line. If we vary c , then we obtain a set or family of horizontal lines

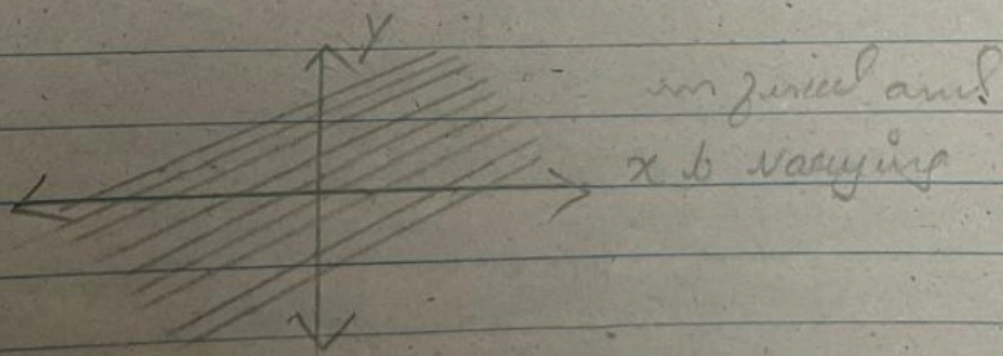


Constants that are varied to produce families of curves are called parameters.

We recall that an equation of the form $y = mx + b$ represents a line of slope m and y -intercept b . If we keep b fixed and treat m as a parameter, then we obtain a family of lines whose members all have y -intercept b .



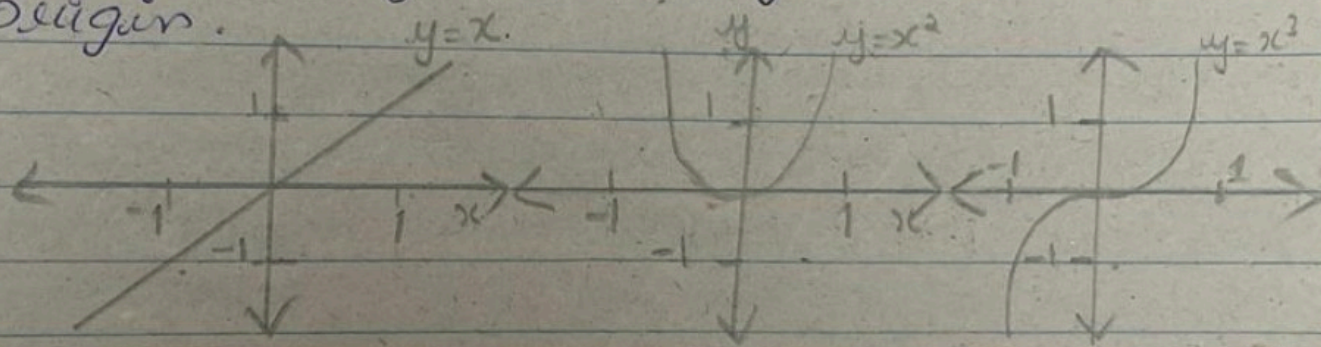
and if we keep m fixed and treat b as a parameter, we obtain a family of parallel lines whose members all have slope m .



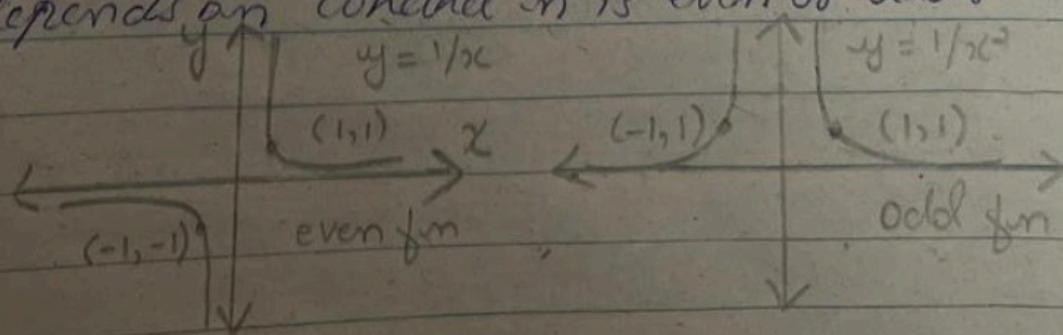
Power Functions;

The Family $y = x^m$:- A function of the form $f(x) = x^p$, where p is ^{the} constant is called a power function. for $m \geq 2$ the shape of the curve $y = x^m$ depends on whether m is even or odd.

- for even values of m , the functions $f(x) = x^m$ or $1/x^m$ are even, so their graphs are symmetric about the y -axis
- for odd values of m , the functions $f(x) = x^m$ or $1/x^m$ are odd, so their graphs are symmetric about the origin.



The Family $y = x^{-n}$:- If p is a negative integer, say $p = -n$, then the power functions $f(x) = x^p$ have the form $f(x) = x^{-n} = 1/x^n$. The shape of the curve depends on whether n is even or odd.



Note: For both even and odd values of n the graph $y = 1/x^n$ has a break at the origin (a discontinuity), which occurs because division by zero is undefined.

Inverse Proportions:-

A variable y is said to be inversely proportional to a variable x if there is a +ve constant k , called the constant of proportionality. Such that

$$(i) - y = \frac{k}{x} \quad \text{or} \quad xy = k \quad - (i)$$

eqn (i) $\Rightarrow$ since k is assumed to be +ve, the graph of (i) has the same shape as $y = 1/x$ but it is compressed or stretched in the y -direction.

eqn (ii) $\Rightarrow$ that the product of inversely proportional variables is a positive constant.

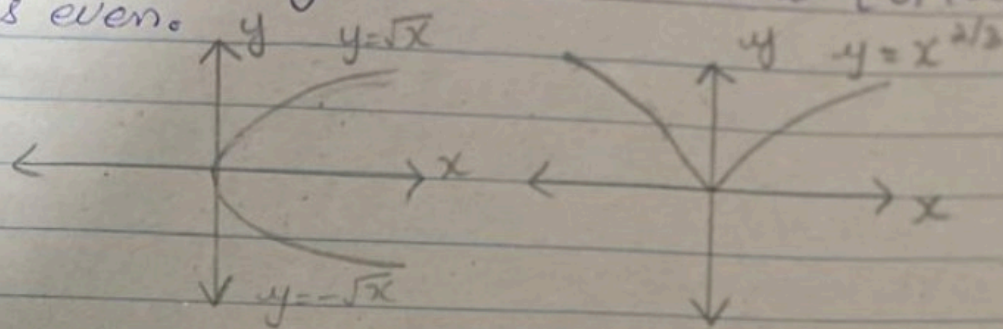
Power Functions With Noninteger Exponents:-

If $P = 1/n$, where n is a positive integer, then the power functions $f(x) = x^P$ have the same form

$$f(x) = x^{1/n} = \sqrt[n]{x}$$

$\because$ every real number has a real cube root, the domain of the function $f(x) = \sqrt[3]{x}$ is $(-\infty, \infty)$ and hence the graph of $y = \sqrt[n]{x}$ extends over the entire x -axis if n is odd.

but extends only over the interval $[0, +\infty)$ if n is even.



Polynomials: A polynomial in x is a function that is expressible as a sum of finitely many terms of the form Cx^n , where C is a constant and n is a non-negative integer. It can be expanded by the binomial formula and expressed as a sum of terms of the form Cx^n . In general it can be written as

$$C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

$$C_nx^n + C_{n-1}x^{n-1} + \dots + C_1x + C_0$$

- * The constants $C_0, C_1, \dots, C_n$ are called the coefficients of the polynomial.
- * The highest power of the x that occurs with a non-zero coefficient is called the degree of the polynomial.
- * Non-zero constant polynomials are considered to have degree 0, i.e. $C = Cx^0$

Polynomials of degree 1, 2, 3, 4 and 5 are described as linear, quadratic, cubic, quartic and quintic respectively.

The natural domain of a polynomial in x is $(-\infty, +\infty)$, the only operations involved are multiplication and addition. The range depends on the particular polynomial.

Rational Functions:- A function that can be expressed as a ratio of two polynomials is called a rational function. If $P(x)$ and $Q(x)$ are polynomials, then the domain of the rational function $f(x) = \frac{P(x)}{Q(x)}$

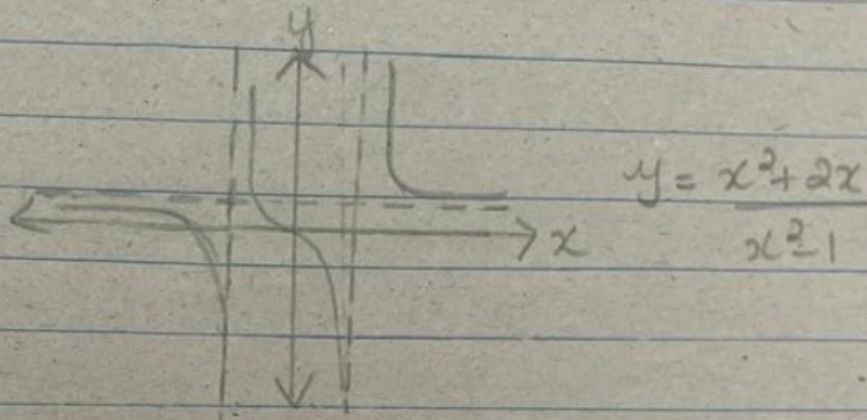
consist of all values of x such that $Q(x) \neq 0$.
For e.g. the domain of the rational function.

$$f(x) = \frac{x^2 + 2x}{x^2 - 1}$$

consist of all values of x , except $x=1$ and $x=-1$.

- The graph of rational functions have discontinuities at the points where the denominator is zero.
- Many rational functions may have graphs that closely approximate a vertical line, called a vertical asymptote. These are represented by the dashed vertical line.

- The graphs of many rational functions eventually get closer and closer to some horizontal line; called a ~~horizontal~~ asymptote, as one traverses the curve in either horizontal the +ve or -ve direction. It is represented by the dashed horizontal lines.



Algebraic Functions :-

Functions that can be constructed from polynomials by applying finitely many algebraic operations (+, -, $\times$, $\div$ and root extraction) are called algebraic functions. e.g. $f(x) = 3\sqrt[3]{x(2+x)}$

The Families $y = A \sin Bx$ and $y = A \cos Bx$

the trigonometric functions of the form

(i) — $f(x) = A \sin(Bx - C)$ and $g(x) = A \cos(Bx - C)$

where A , B and C are nonzero constants.

The graphs of such functions can be obtained by stretching, compressing, translating

and reflecting the graphs of

$$y = \sin x \text{ and } y = \cos x \text{ — (ii)}$$

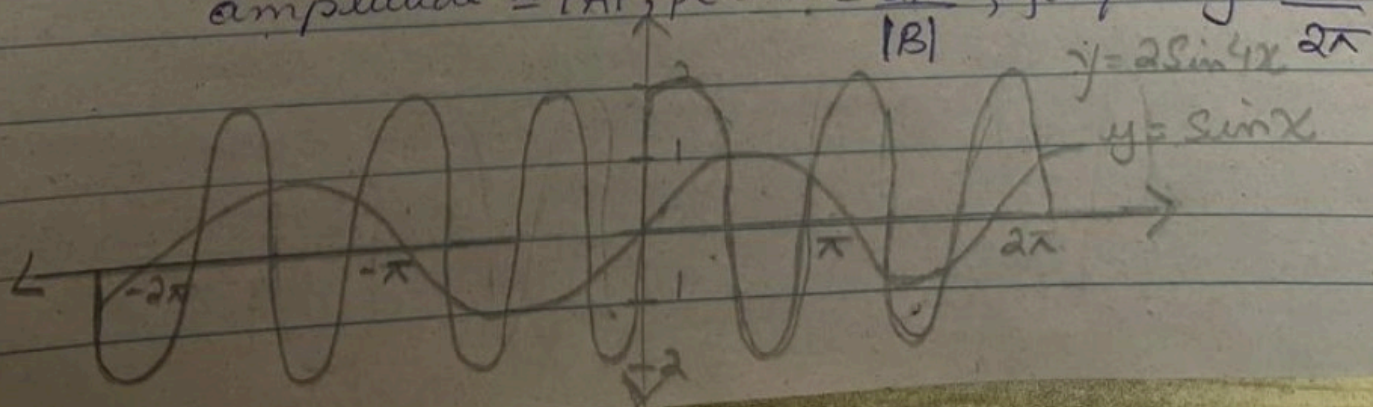
Case 1:- where $C = 0$

eqn (i) $\Rightarrow y = A \sin Bx$ and $y = A \cos Bx$ — (iii)
related to the graphs of eqn (ii).

- * The effect of constant A is to stretch or compress the graphs of eqn (ii) vertically.
- * The effect of constant B is to stretch or compress the graphs of eqn (ii) horizontally.

Note: The multiplier of x stretches when it is less than 1 and compresses when it is greater than 1.

- * If A and B are +ve numbers, then the graphs of eqn (iii) oscillates ~~between~~ between $-A$ and A and repeat every $2\pi/B$ units at amplitude A , $f = \frac{B}{2\pi}$.
- * If these constants are -ve numbers, then these constant cause reflections of the graphs about the axes as well as compressing or stretching them and in this case
amplitude $= |A|$, period $= \frac{2\pi}{|B|}$; frequency $= \frac{|B|}{2\pi}$



The Families $y = A \sin(Bx - C)$ and $y = A \cos(Bx - C)$
 To investigate the graphs of the more
 general families

$y = A \sin(Bx - C)$ and $y = A \cos(Bx - C)$
 can be rewritten as:

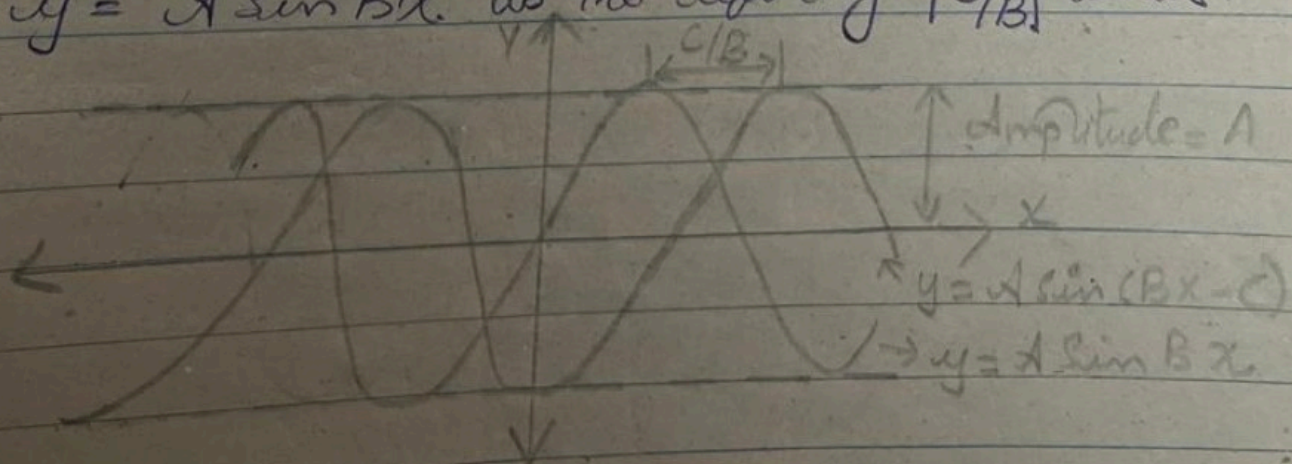
$$y = A \sin\left[B\left(x - \frac{C}{B}\right)\right] \text{ and } y = A \cos\left[B\left(x - \frac{C}{B}\right)\right]$$

Thus the graphs of these eqns can be
 obtained by translating the graph of
 $y = A \sin Bx$ and $y = A \cos Bx$ to the left
 or right, depending on the sign of C/B . If $\frac{C}{B} > 0$
 then the graph of

$$y = A \sin\left[B\left(x - \frac{C}{B}\right)\right]$$

can be obtained by translating the graph
 of $y = A \sin Bx$ to the right by $\frac{C}{B}$ units.

If $\frac{C}{B} < 0$, the graph of $y = A \sin(Bx - C)$
 is obtained by translating the graph
 of $y = A \sin Bx$ to the left by $|C/B|$ units.



In mathematics the term inverse is used to describe functions that reverse one another in the sense that each undoes the effect of the other.

Inverse Functions :-

The idea of solving an equation $y = f(x)$ for x as a function of y , say $x = g(y)$ is one of the most important ideas in mathematics. For eg.

$y = x^3 + 1 \Rightarrow y = f(x)$ — (i)
can be solved for x as a function of y .

$x = \sqrt[3]{y-1} \Rightarrow x = g(y)$ — (ii)
from eqn (i) we can compute y if x is known
from eqn (ii) we can compute x if y is known.

On this we have to identify the relationships that exist between f and g from eqn. (i) and (ii)

Definition :- If the functions f and g satisfy the two conditions

$$g(f(x)) = x \quad \forall x \text{ in the domain of } f$$

$$f(g(y)) = y \quad \forall y \text{ in the domain of } g$$

then, f is a inverse of g and
 g is a unverse of f or
 f and g are inverse functions.

Changing the independent variable :-

$$f^{-1}(f(x)) = x \quad \forall x \text{ in the domain of } f$$

Here x use as the independent variable of f .

$$f(f^{-1}(y)) = y \quad \forall y \text{ in the domain of } f^{-1}$$

Here y use as the independent variable of f^{-1} .

These are the cancellation equations for f and f^{-1} .

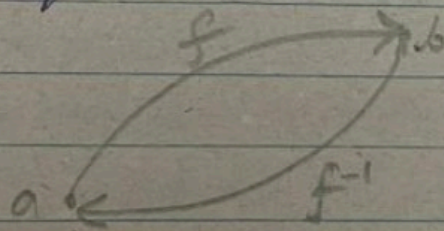
Domain and Range of Inverse Functions :-

The relation-ships between the domain and ranges of f and f^{-1} .

$$\text{domain of } f^{-1} = \text{range of } f$$

$$\text{range of } f^{-1} = \text{domain of } f$$

These eqns are subset of each other.



if f maps a to b , then f^{-1} maps b to a

A Procedure For Finding The Inverse of A Function f :-

- i) Write down the eqn $y = f(x)$
- ii) If possible, solve this eqn for x as a function of y .
- iii) The resulting eqn will be $x = f^{-1}(y)$, which provides a formula for f^{-1} with y as the independent variable.

iv) If y is acceptable as the independent variable for the inverse function, then you are done, but if you want to have x as the independent variable, then you need to interchange x and y in the equation $x = f^{-1}(y)$ to obtain $y = f^{-1}(x)$

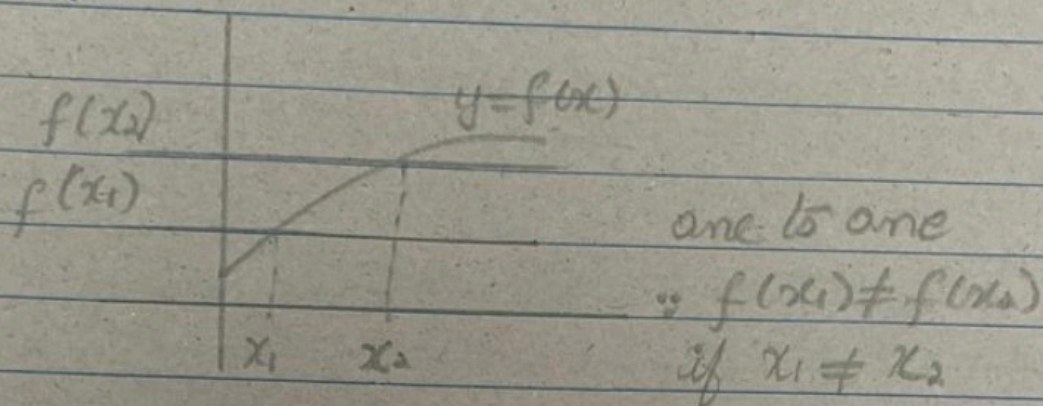
Existence of Inverse Functions :

If a function f has an inverse, then it must assign distinct outputs to distinct inputs, for e.g. i.e. one to one or invertible
for e.g. $f(x) = x^2$ cannot have an inverse because it assigns the same value to $x=2$ and -2
i.e. $f(2) = f(-2) = 4 \Rightarrow f^{-1}(4) = 2$ and $f^{-1}(4) = -2$
which is not possible.

also; if $y = x^2$ you x as a function of y
 $\Rightarrow x = \pm\sqrt{y}$ does not express x as a single function of y .

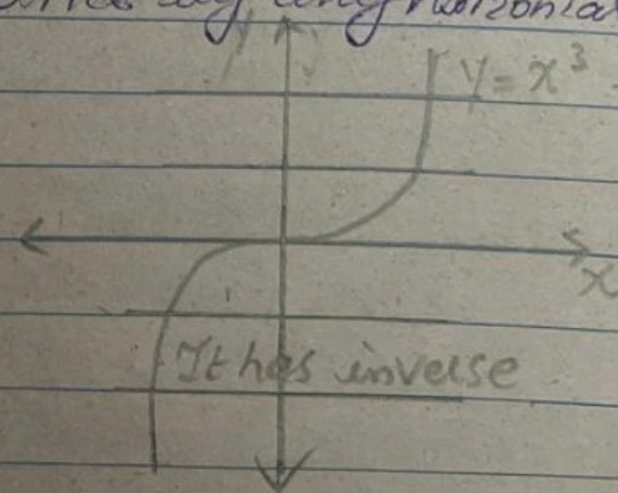
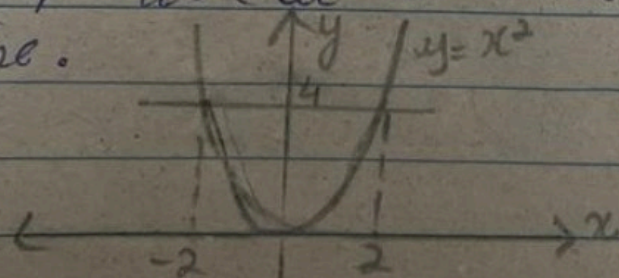
Theorem 1: If an equation $y = f(x)$ can be solved for x as a function of y , say $x = g(y)$, then f has an inverse and that inverse is $g(y) = f^{-1}(y)$.

Theorem 2: A function has an inverse iff it is one to one. i.e. $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.



Theorem 3: (The horizontal line test)

A function has an inverse function iff its graph is cut at most once by any horizontal line.



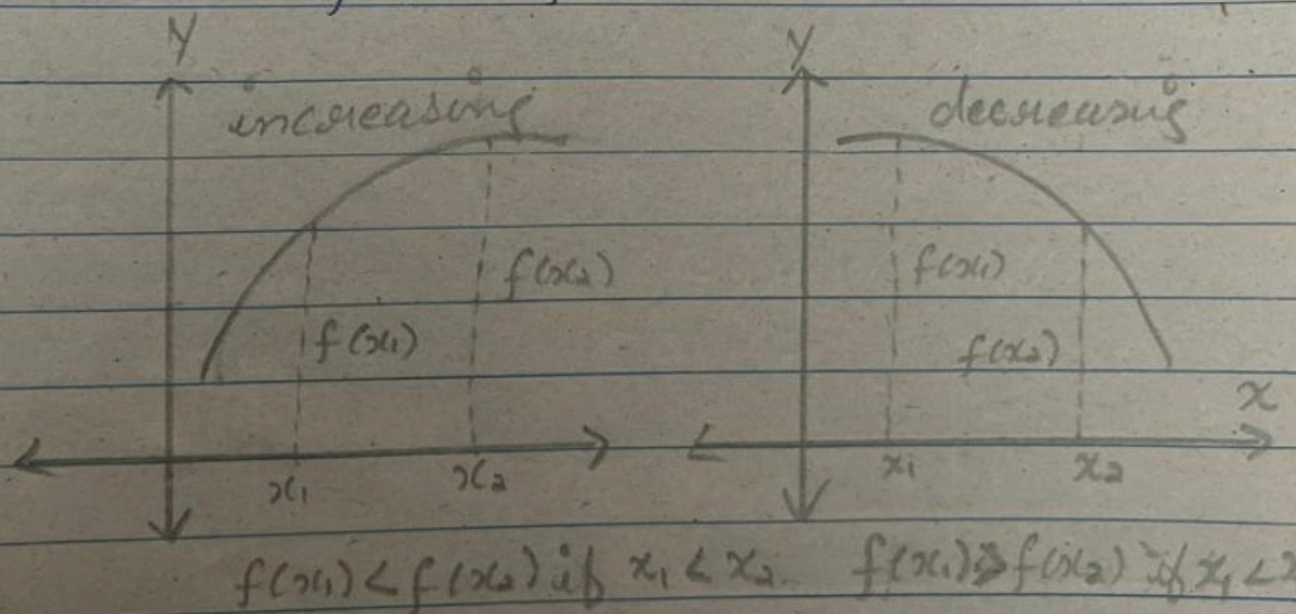
Increasing Or Decreasing Functions Are Invertible :-

A function whose graph is always rising as it is traversed from left to right is said to be an increasing function. i.e.

~~Ex~~ $f(x_1) < f(x_2)$ whenever $x_1 < x_2$

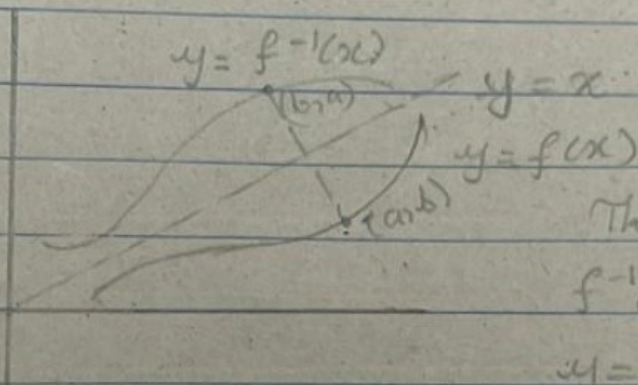
A function whose graph is always falling as it is traversed from left to right is said to be decreasing function. If x_1 and x_2 are points in the domain of a function f then f is decreasing if

$$f(x_1) > f(x_2) \text{ whenever } x_1 < x_2$$



Graphs of Inverse Function :-

Theorem 4.1- If f has an inverse, then the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about the line $y = x$; i.e. each graph is the mirror image of the other w.r.t that line.



The graphs of f and f^{-1} are reflections about $y = x$.

Restricting Domains Has Invertibility

If a function g is obtained from a function f by placing restrictions on the domain of f , then g is called a restriction of f .
Thus, for e.g., the function

$$g(x) = x^3 \quad x \geq 0$$

is a restriction of the function $f(x) = x^3$.
It is called the restriction of x^3 to the interval $[0, +\infty)$

Inverse Trigonometric Functions -

The six basic trigonometric functions do not have inverses because their graphs repeat periodically and hence do not pass the horizontal line test. To overcome this problem we will restrict the domains of the trigonometric functions to produce one-to-one functions and define them as "inverse trigonometric functions".

The following formal definitions summarize the preceding discussion.

1. The inverse sine function, denoted by $\sin^{-1}$, is defined to be the inverse of the restricted sine function. $\sin x, -\pi/2 \leq x \leq \pi/2$
2. The inverse cosine function, denoted by $\cos^{-1}$, is defined to be the inverse of the restricted cosine function. $\cos x, 0 \leq x \leq \pi$
3. The inverse tangent function, denoted by $\tan^{-1}$, is defined to be the inverse of the restricted tangent function. $\tan x, -\pi/2 < x < \pi/2$
4. The inverse secant function, denoted by $\sec^{-1}$, is defined to be the inverse of the restricted secant function. $\sec x, 0 \leq x \leq \pi$ with $x \neq \pi/2$

Note: A reflection about $y=x$ converts vertical lines into horizontal lines, and vice versa, and it converts x -intercepts into y -intercepts, and vice versa.

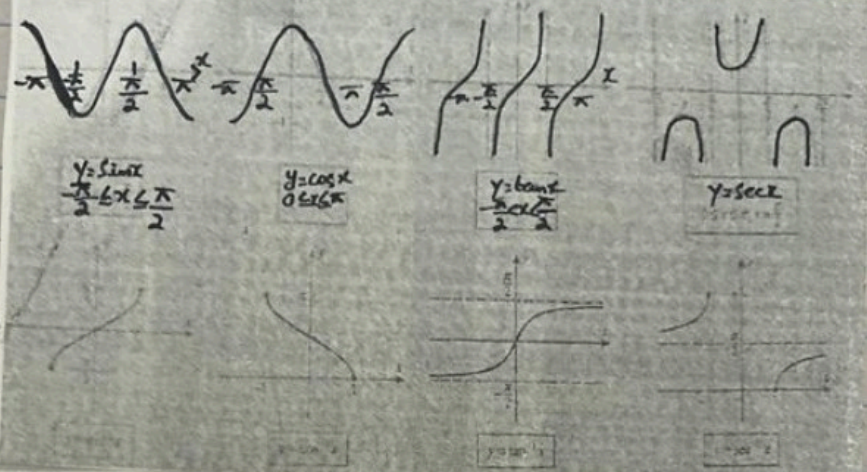


Table 0.4.1

PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

FUNCTION	DOMAIN	RANGE	BASIC RELATIONSHIPS
$\sin^{-1}$	$[-1, 1]$	$[-\pi/2, \pi/2]$	$\sin^{-1}(\sin x) = x$ if $-\pi/2 \leq x \leq \pi/2$ $\sin(\sin^{-1} x) = x$ if $-1 \leq x \leq 1$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$	$\cos^{-1}(\cos x) = x$ if $0 \leq x \leq \pi$ $\cos(\cos^{-1} x) = x$ if $-1 \leq x \leq 1$
$\tan^{-1}$	$(-\infty, +\infty)$	$(-\pi/2, \pi/2)$	$\tan^{-1}(\tan x) = x$ if $-\pi/2 < x < \pi/2$ $\tan(\tan^{-1} x) = x$ if $-\infty < x < +\infty$
$\sec^{-1}$	$(-\infty, -1] \cup [1, +\infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$	$\sec^{-1}(\sec x) = x$ if $0 \leq x < \pi/2$ or $\pi/2 < x \leq \pi$ $\sec(\sec^{-1} x) = x$ if $ x \geq 1$

Exponential And Logarithmic Functions

Rational Exponents: If b is a nonzero real number, then non zero integer powers of b are defined by

$$b^n = b \times b \times \dots \times b \text{ and } b^{-n} = \frac{1}{b^n}$$

and if $n=0$, then $b^0 = 1$. Also, if P/Q is a rational number expressed in lowest terms then

$$b^{P/Q} = \sqrt[Q]{b^P} = (\sqrt[Q]{b})^P \text{ and } b^{-P/Q} = \frac{1}{b^{P/Q}}$$

where we assume $b > 0$

If b is negative, then some rational powers of b will have imaginary values such as $2^{\sqrt{2}}$, $3^{\sqrt{2}}$, 1^{-7}

Laws of Exponents

i. $b^p b^q = b^{p+q}$ ii. $\frac{b^p}{b^q} = b^{p-q}$ iii. $(b^p)^q = b^{pq}$

iv. $b^0 = 1$

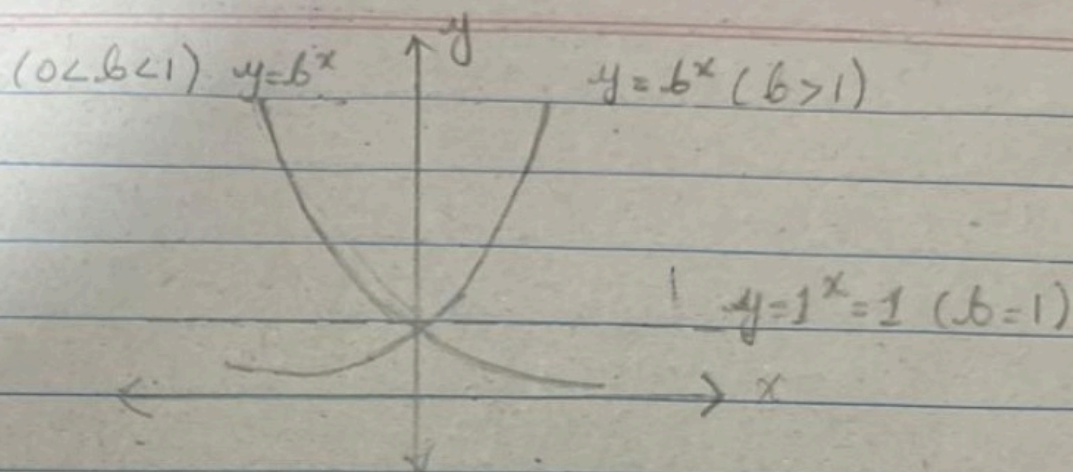
The Family of Exponent Functions :-

A function of the form $f(x) = b^x$, where $b > 0$, is called an exponential function with base b . For e.g. $f(x) = 2^x, (1/2)^x, \pi^x$

An exponential function has a constant base and variable exponent.

The graph of $y = b^x$ has the following properties:-

- The graph passes through $(0, 1)$ because $b^0 = 1$
 - If $b > 1$, the value of b^x increases as x increases. As you traverse the graph from left to right, the value of b^x increases indefinitely. If you traverse the graph from right to left the value of b^x decreases toward zero but never reach zero. Thus the x -axis is a horizontal asymptote of the graph of b^x .
 - If $0 < b < 1$, the value of b^x decreases as x increases. The next will be vice versa of condition point 2.
 - If $b = 1$, then the value of b^x is constant.
- The graph of $y = (1/b)^x$ is the reflection of the graph of $y = b^x$ about the y -axis. Because of replacing x by $-x$ in the equation
- $$y = b^x \Rightarrow y = b^{-x} \Rightarrow y = (1/b)^x$$



The Natural Exponential Function :-

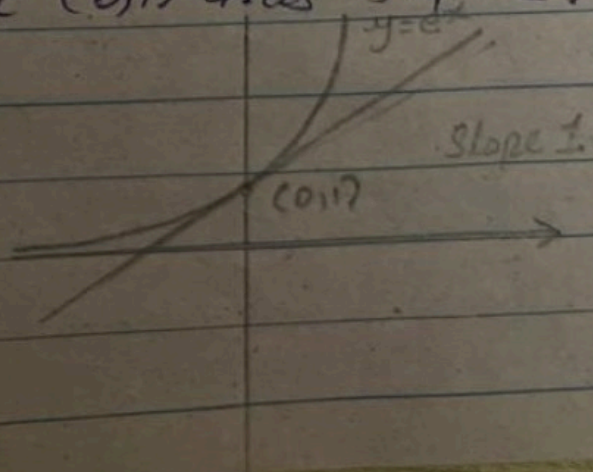
The function $f(x) = e^x$ is called the natural exponential function. It is also denoted as $\exp(x)$, in which case the relationship $e^{x_1+x_2} = e^{x_1}e^{x_2}$ or

$$\exp(x_1+x_2) = \exp(x_1)\exp(x_2)$$

That base, denoted by the letter e , is a certain irrational number whose value to six decimal places is

$$e \approx 2.718282$$

Also, $b=e$ is the only base for which the slope of the tangent line to the curve $y=b^x$ at any point P on the curve is equal to y -coordinate at P . Thus, the tangent line to $y=e^x$ at $(0,1)$ has slope 1.



Logarithmic Functions

The function $f(x) = \log_b x$ the logarithmic function with base b . Logarithmic functions can also be viewed as inverses of exponential functions. If $b > 0$ and $b \neq 1$, then the graph of $f(x) = b^x$ passes the horizontal line test so b^x has an inverse. Formula for this inverse with x as the independent variable is $x = b^y$ for y as a function of x and can be written as $y = \log_b x$.

Theorem 2- If $b > 0$ and $b \neq 1$, then b^x and $\log_b x$ are inverse functions.

It follows that the graphs of $y = b^x$ and $y = \log_b x$ are reflections of one another about the line $y = x$. In general,

$$y = \ln x \text{ iff } x = e^y$$

Property of b^x	Property of $\log_b x$	Formulae
$b^0 = 1$	$\log_b 1 = 0$	$\log_b(ac) = \log_b a + \log_b c$
$b^1 = b$	$\log_b b = 1$	$\log_b(a/c) = \log_b a - \log_b c$
Range is $(0, +\infty)$	Domain is $(0, +\infty)$	$\log_b(a^x) = x \log_b a$
Domain is $(-\infty, +\infty)$	Range is $(-\infty, +\infty)$	$\log_b(1/c) = -\log_b c$
x -axis is a horizontal asymptote	y -axis is a vertical asymptote	